Class 2

Make It Yours

Workspace, Skills, and Automations

Quick Check: Class 1 Recap

You should have:

- Ubuntu running on your dedicated laptop
- OpenClaw installed with gateway daemon running
- Telegram connected and responding to messages

If anything broke since last class, open a terminal and run:

```
npx openclaw gateway restart
npx openclaw doctor
```

If you need help, raise your hand now — we'll fix it before moving on.

What We'll Do Today

By the end of this class, you will have:

1. **Configured** your agent's workspace files (who it is, who you are)
2. **Installed** useful skills from ClawHub
3. **Set up** a daily morning briefing on Telegram
4. **Understood** context engineering — how to make your agent smarter

The Workspace: Your Agent's Brain

Everything your agent knows lives in `~/.openclaw/workspace/`

```
ls ~/.openclaw/workspace/
```

The 9 Core Files

File	What It Does
IDENTITY.md	Agent's name, role, purpose
SOUL.md	Personality, communication style
USER.md	Everything about YOU (the owner)
MEMORY.md	Long-term facts and learnings (grows over time)
AGENTS.md	Operating rules and boundaries
HEARTBEAT.md	Scheduled task instructions
BOOT.md	Runs every gateway start
BOOTSTRAP.md	Runs on first boot only
TOOLS.md	API tools documentation

How Workspace Files Work

Hot Reload:

Edit any `.md` file in the workspace → changes take effect on the **next message**. No restart needed.

File Size Limits:

- **20,000 characters** per file (max)
- **150,000 characters** total across all files
- Files over the limit are silently truncated — keep them concise

The Rule:

> ***Write for the machine, not for yourself.*** *Use headings, bullets, and tables. Skip prose. Be specific.*

USER.md — Teaching Your Agent About You

This is the most important file. It's your agent's permanent knowledge of who you are.

What to Include:

- Your name, location, timezone
- What you do for work
- Current goals (next 6 months)
- Preferences (communication style, tools you use)
- Important people in your life (names, relationships)
- Daily routine and habits
- What you're learning or working on

> *Think of it as the brain dump you'd give a new executive assistant on day one.*

Exercise: Write Your USER.md (15 min)

```
nano ~/.openclaw/workspace/USER.md
```

Write at least these sections:

```
## About Me        → Name, location, what you do
## Goals           → 3-5 goals for the next 6 months
## Preferences     → Communication style, tools you use
## Daily Routine   → Wake time, key daily activities
```

Save: Ctrl+O, Enter. Exit: Ctrl+X.

Changes take effect immediately — test by messaging your agent on Telegram.

SOUL.md — Your Agent's Personality

This defines **how** your agent communicates, not what it knows.

What to Include:

- Communication style (formal, casual, direct, gentle)
- How it should push back on bad ideas
- When to ask for clarification vs. just do it
- Hard rules ("never do X", "always do Y")

SOUL.md — Example

```
## Communication Style
Direct and concise. No fluff. Lead with yes or no.

## Hard Rules
- Never send messages without my explicit approval
- If unsure about a destructive action, ask first
- Keep responses under 3 paragraphs unless I ask for more
```

Exercise: Write Your SOUL.md (10 min)

```
nano ~/.openclaw/workspace/SOUL.md
```

Write at least a communication style and hard rules section.

Then test it: send your agent a message on Telegram and see if the tone matches what you wrote.

Pro tip: you can also just tell your agent in chat — "Update your SOUL.md with these rules: ..." — and it will edit the file itself.

IDENTITY.md — Naming Your Agent

```
nano ~/.openclaw/workspace/IDENTITY.md
```

Keep it simple:

```
## Name        → [Your agent's name]
## Role        → Personal AI assistant for [your name]
## Purpose     → Help me manage my daily life and get things done
```

Pick a name that feels right. Human names, robot names, mythology — it's your agent.

Context Engineering Tips

Principle	Example
Be specific	"I live in Jersey City, NJ" not "I live near NYC"
Use structure	Headings, bullets, tables — not long paragraphs
Verbalize the implicit	Write down things you take for granted
Update monthly	Goals change. Routines shift. Keep files current.
Less is more	20K char limit exists for a reason — cut the fluff

> ***Every word in your workspace costs tokens.*** *Make them count.*

What Are Skills?

Skills are **markdown instruction files** that teach your agent new capabilities.

- Stored in `~/.openclaw/workspace/skills/`
- No code — just text-based instructions with YAML metadata
- Thousands of community skills on **ClawHub** marketplace

Examples:

- Weather lookups, news summaries, calendar integration
- Obsidian note-taking, email management, web scraping
- Crypto tracking, fitness logging, recipe finding

ClawHub Security: The 1,184 Malicious Skills

In 2026, security researchers found **1,184 malicious skills** on ClawHub.

- One attacker uploaded **677 packages** alone
- Disguised as crypto bots, YouTube tools, wallet trackers
- The **#1 ranked skill** had 9 vulnerabilities
- Payloads: password stealers, reverse shells, data exfiltration

Before installing ANY skill:

1. Run `npx openclaw skills info [name]` and **read the output**
2. Look for: excessive permissions, external server calls, security-disabling instructions
3. Pin the version: `npx clawhub install [name] --pin`

Exercise: Install Skills (15 min)

Search, install, and test:

```
npx clawhub search weather        # or: news, calendar, etc.
npx clawhub install SKILL-NAME --pin
npx openclaw gateway restart
```

Then message your agent on Telegram and try a trigger phrase.

Important:

- Install **1-2 skills max** — each uses context window space
- 50 skills = 25,000 tokens before you even say a word
- Pin the version (`--pin`) to prevent silent updates

Managing Skills

```
npx openclaw skills list                                    # List installed
npx openclaw config set skills.entries.NAME.enabled false    # Disable (saves tokens)
npx clawhub delete SKILL-NAME                                # Remove completely
npx openclaw gateway restart                                 # Apply changes
```

> ***Rule of thumb:*** *Only keep skills you use at least weekly. Disable the rest.*

Disabled skills stay installed but don't consume context window tokens.

What Are Cron Jobs?

Scheduled tasks that run automatically at specific times.

Schedule	Meaning
`0 7 * * *`	7:00 AM every day
`30 8 * * 1-5`	8:30 AM weekdays only
`0 9 * * 1`	9:00 AM every Monday
`0 */2 * * *`	Every 2 hours

Format: `minute hour day-of-month month day-of-week`

The most common use: **morning briefings** via Telegram.

The Morning Brief

Tell your agent to send you a daily summary every morning:

What a good morning brief includes:

1. Date and personalized greeting
2. Weather for your city
3. Top 3 news stories in your interest areas
4. Today's tasks and reminders
5. One proactive suggestion based on your goals

The easiest way to set it up:

> *Just message your agent: "Set up a daily morning brief at 7 AM on Telegram. Include weather for [your city], top AI news, my tasks, and one suggestion."*

Exercise: Set Up Your Morning Brief (10 min)

Method 1: Just ask your agent (easiest)

Tell it in Telegram what you want and when.

Method 2: Config commands (precise control)

```
npx openclaw config set agents.defaults.heartbeat.every "0 8 * * *"
npx openclaw config set agents.defaults.heartbeat.model "claude-sonnet-4-6"
npx openclaw config set agents.defaults.heartbeat.target "telegram"
npx openclaw gateway restart
```

Morning Brief: Active Hours

Prevent 3 AM pings:

```
npx openclaw config set agents.defaults.heartbeat.activeHours.start "07:00"
npx openclaw config set agents.defaults.heartbeat.activeHours.end "23:00"
npx openclaw config set agents.defaults.heartbeat.activeHours.timezone "America/New_York"
```

Replace the timezone with yours. This ensures heartbeats only fire during waking hours.

Heartbeat Cost Tips

Model	Cost per Heartbeat	Monthly (daily)
Claude Opus 4.6	~$0.05-0.15	~$2-5
Claude Sonnet 4.6	~$0.01-0.05	~$0.50-1.50
Claude Haiku 4.5	~$0.001-0.005	~$0.03-0.15

Save money:

- Use **Haiku or Sonnet** for heartbeats (Opus for real conversations)
- Set heartbeat interval to **55 minutes** — hits Anthropic's prompt cache (90% savings)
- **activeHours** = zero tokens wasted overnight

The Three Essential Automations

Automation	Schedule	What It Does
Morning Brief	Daily at your wake time	News, weather, tasks, suggestions
Workspace Backup	Every few hours	`cp -r ~/.openclaw/workspace ~/backup-$(date +%Y%m%d)`
Security Audit	Weekly	`npx openclaw security audit --deep --fix`

Set up the morning brief today. Backups and security audits are covered in Class 3.

Verify Everything Works

Test each piece:

1. **Workspace:** Message your agent — "What do you know about me?" *(Should reflect your USER.md)*
2. **Personality:** "Tell me a joke" *(Should match your SOUL.md tone)*
3. **Skills:** Try a trigger phrase for your installed skill
4. **Heartbeat:** `npx openclaw config get agents.defaults.heartbeat` *(Should show your schedule)*

All working? You're done with Class 2.

What You Built Today

- **Workspace configured** — your agent knows who you are and how to behave
- **Skills installed** — new capabilities from ClawHub
- **Morning brief** — automated daily summary on Telegram
- **Context engineering** — you know how to make your agent smarter over time

Homework Before Class 3:

- **Refine** your USER.md and SOUL.md based on how the agent responds
- **Use the morning brief** for a few days — adjust the content
- Think about: What should your agent NOT be allowed to do?

Class 3: Lock It Down

Security Hardening and Advanced Features

> *Next time: Docker sandboxing, tool policies, model switching, and maintenance.*